Lecture 7: Models for Human-Robot Collaboration (Part-1)

COMP3018/COMP5018: Human-Robot Interaction

Dr. Amir Aly

School of Engineering, Computing, and Mathematics

Plymouth University

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amir.aly@plymouth.ac.uk

Introduction

- **Today's Topics**

1- Introduction to POMDP

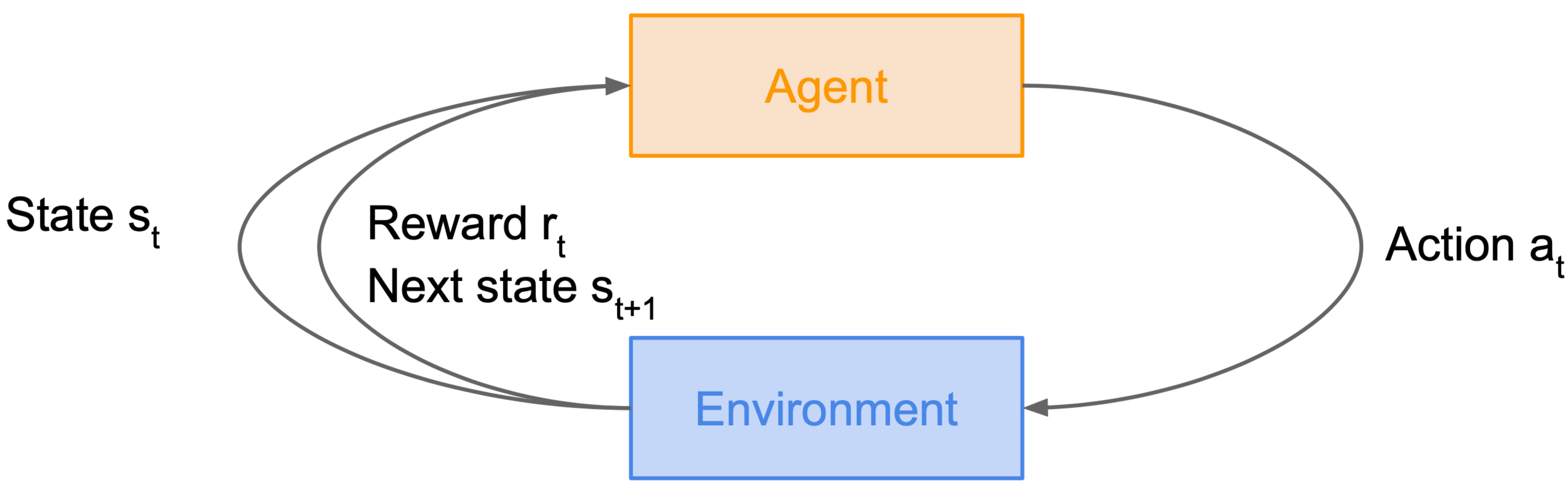
- **Session Learning Outcomes**

By the end of today's lecture you will be able to:

1- Describe the different between MDP and POMDP

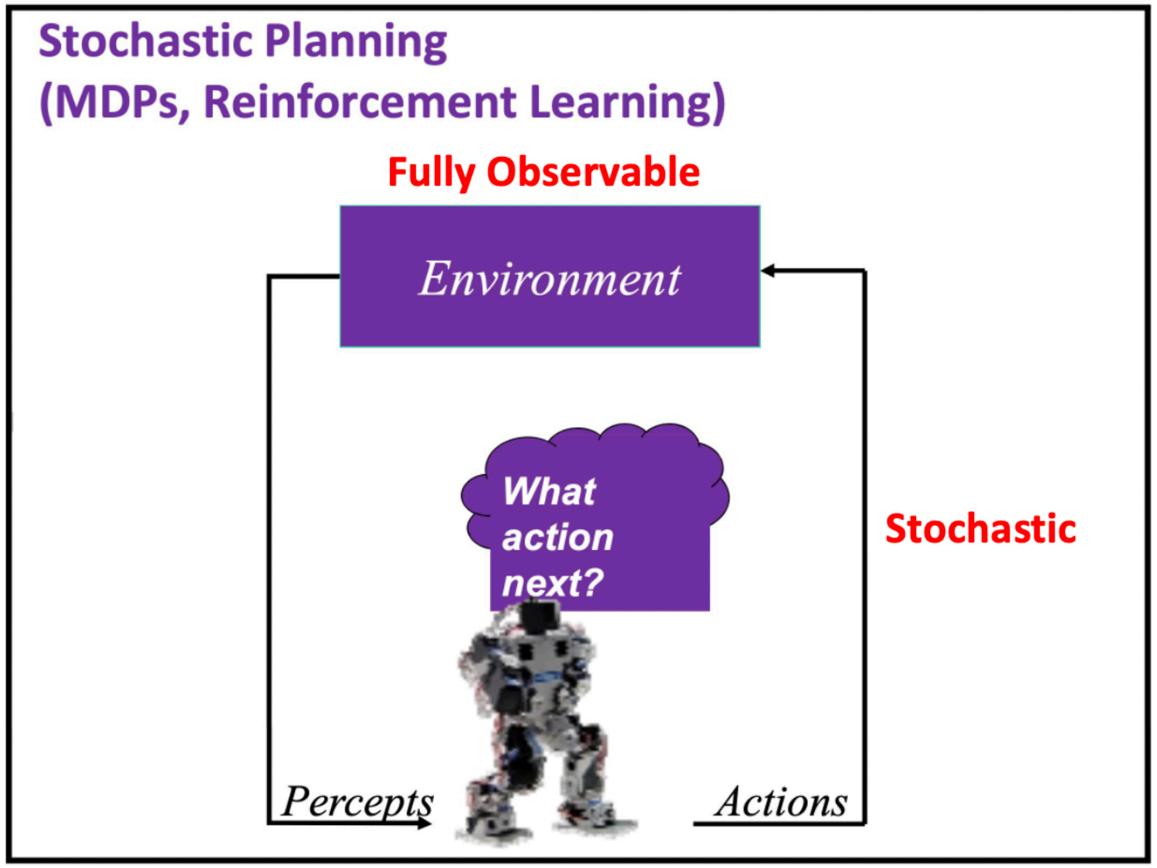
RECALL
Markov Decision Process (MDP)

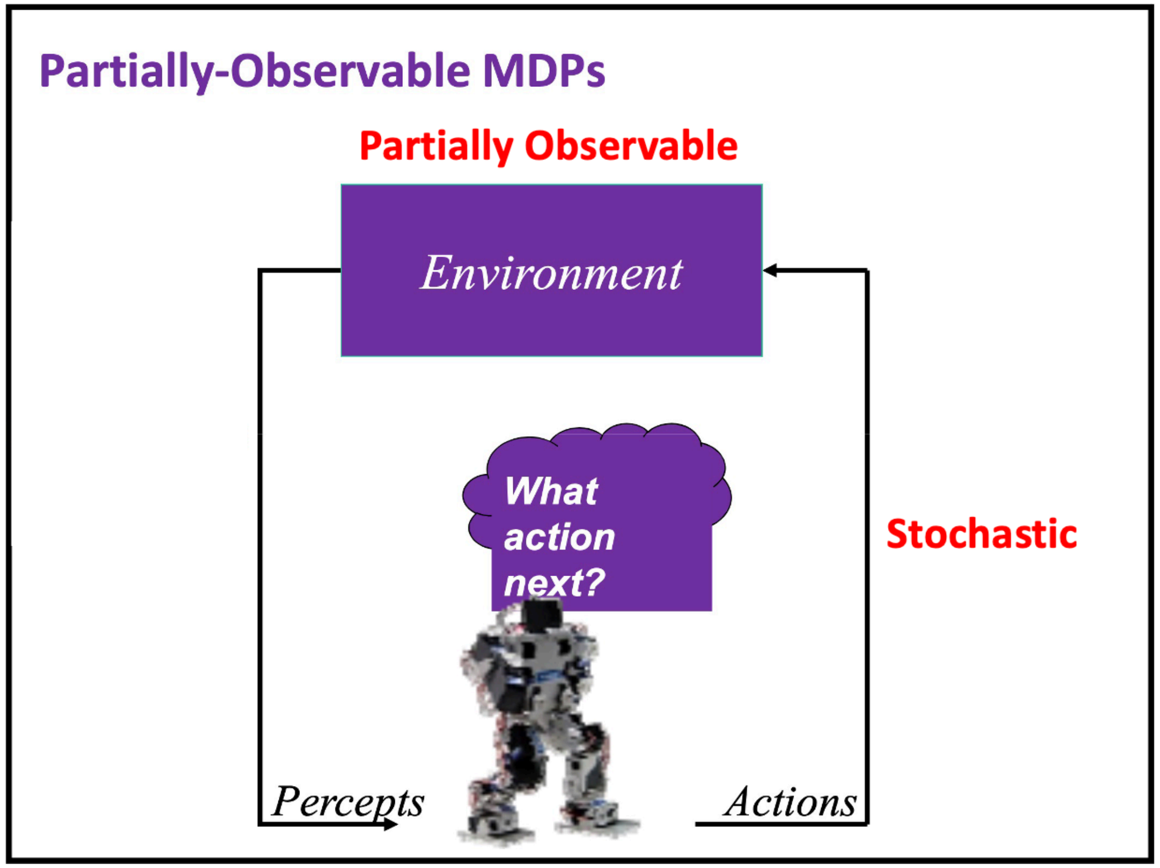
- Problems involving an agent interacting with an environment, which provides numeric reward signals.
- Goal: Learn how to take actions in order to maximize reward.



Partially Observable Markov Decision Process (POMDP)

- A POMDP is really just an MDP; we have a set of states, a set of actions, transitions, and immediate rewards. The actions' effects on the state in a POMDP are exactly the same as in an MDP. The only difference is in whether or not we can observe the current state of the process.





- In an MDP, while you know the state of the environment fully, the outcome of your actions is probabilistic, not deterministic. The outcome of actions in POMDPs is also affected by probabilistic transitions, similar to MDPs, but with the added complexity of uncertainty about the state itself.

Partially Observable Markov Decision Process (POMDP)

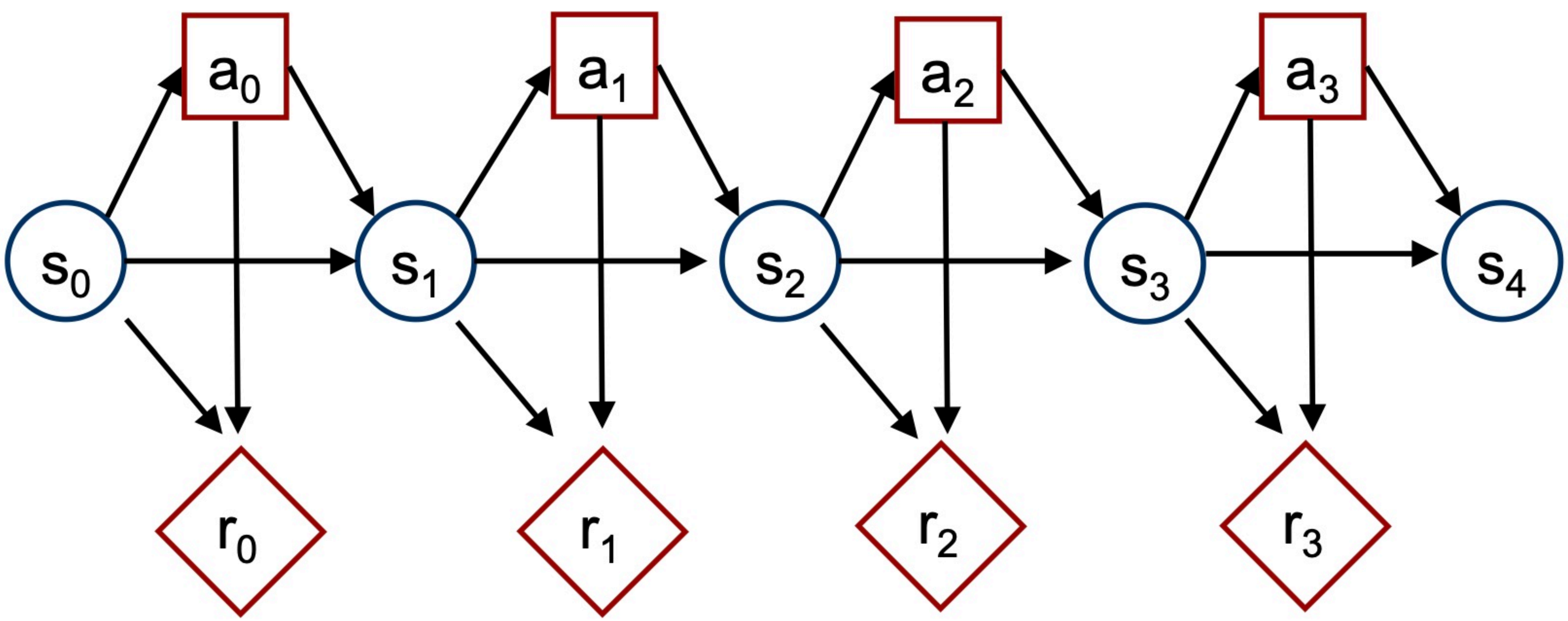
POMDP: UNCERTAINTY

Case #1: Uncertainty about the action outcome

Case #2: Uncertainty about the world state due to imperfect (partial) information

Partially Observable Markov Decision Process (POMDP)

- Fully Observable MDP **(Graphical Model)**



RECALL
Markov Decision Process (MDP)

- **Mathematical formulation of the RL problem**
- **Markov property**: Current state completely characterizes the state of the world

Markov Property

- **The future is independent of the past given the present.**
- **A state S_t is Markov if and only if:**

$$\mathbb{P}[S_{t+1} \mid S_t] = \mathbb{P}[S_{t+1} \mid S_1, \ldots, S_t]$$

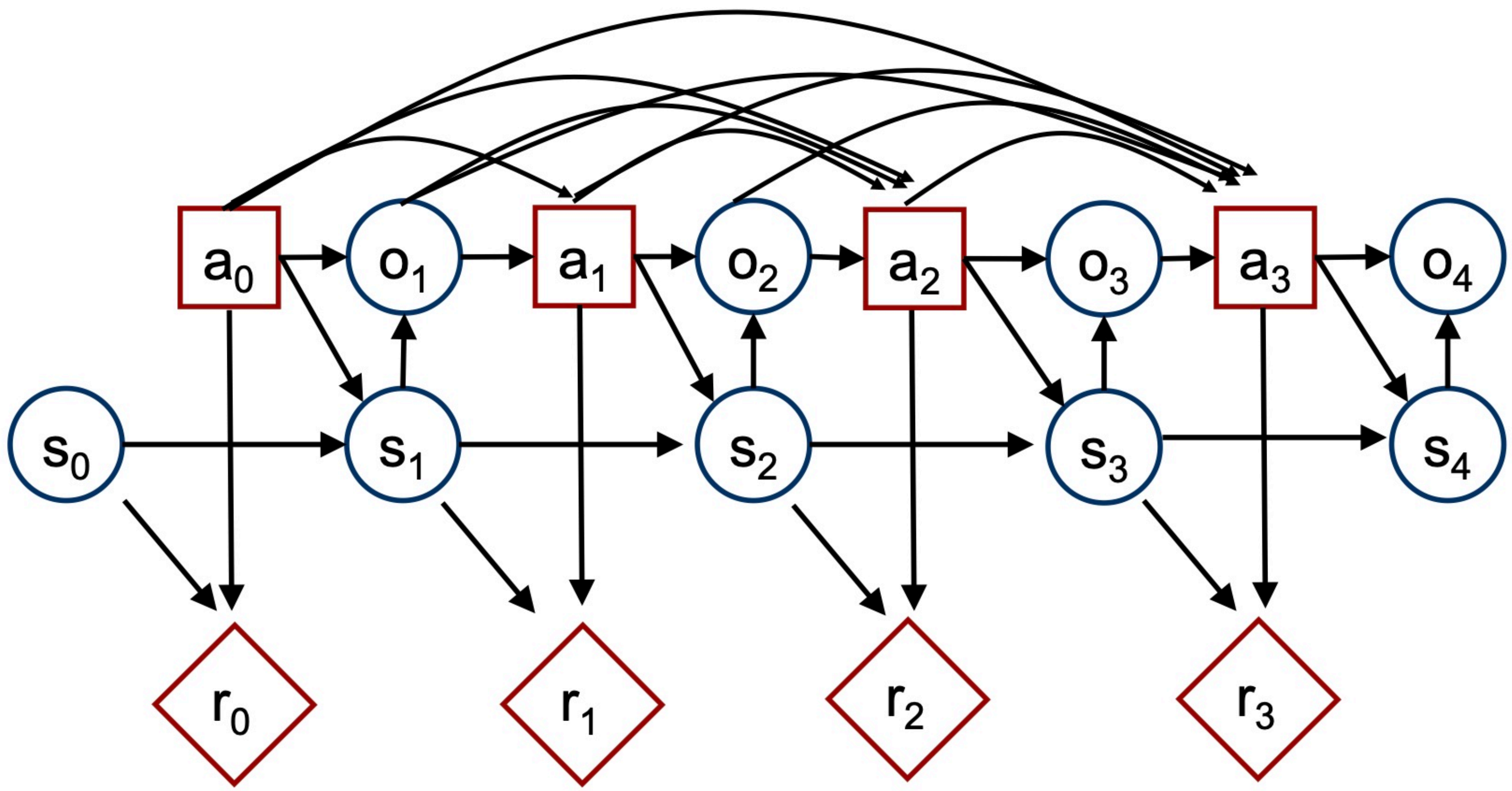
- **The present state captures all relevant information from the history.**

Defined by: $(\mathcal{S}, \mathcal{A}, \mathcal{R}, \mathbb{P}, \gamma)$

$\mathcal{S}$: set of possible states
$\mathcal{A}$: set of possible actions
$\mathcal{R}$: distribution of reward given (state, action) pair
$\mathbb{P}$: transition probability i.e. distribution over next state given (state, action) pair
γ : discount factor, **where** $0 \leq \gamma < 1$ **(For scaling rewards)**

Partially Observable Markov Decision Process (POMDP)

- Partially Observable MDP **(Graphical Model)**



- **The connection between actions and observations at time *t* to future actions in the graphical model for POMDPs represents the <u>dependency of future decisions on past actions and observations</u>. The decision-making process at any time point in a POMDP takes into account the <u>entire history of what has occurred up to that point</u>, which includes past actions and observations, as they inform the belief about the current state and influence future decisions.**

Partially Observable Markov Decision Process (POMDP)

- In POMDPs we apply the very same idea as in MDPs.
- Since the state is not observable, the agent has to make its decisions based on the belief state which is a posterior distribution over states.

- **S**: set of states
- **A**: set of actions
- **Pr**($s'|s,a$): transition model
- **R**(s,a,s'): reward model
- γ: discount factor
- s_0: start state
- **E** set of possible evidence (observations)
- **Pr**($e|s$)

Partially Observable Markov Decision Process (POMDP)

Policies

- MDP policies: $\pi: S \rightarrow A$
  - Markovian policy

The action A to be taken is determined solely by the current state S

- But state is unknown in POMDPs

- POMDP policies: $\pi: B_0 \times H_t \rightarrow A_t$
  - B_0 is the space of initial beliefs b_0

$$b_0 = \Pr(s_0)$$

  - H_t is the space histories h_t of observables up to time t

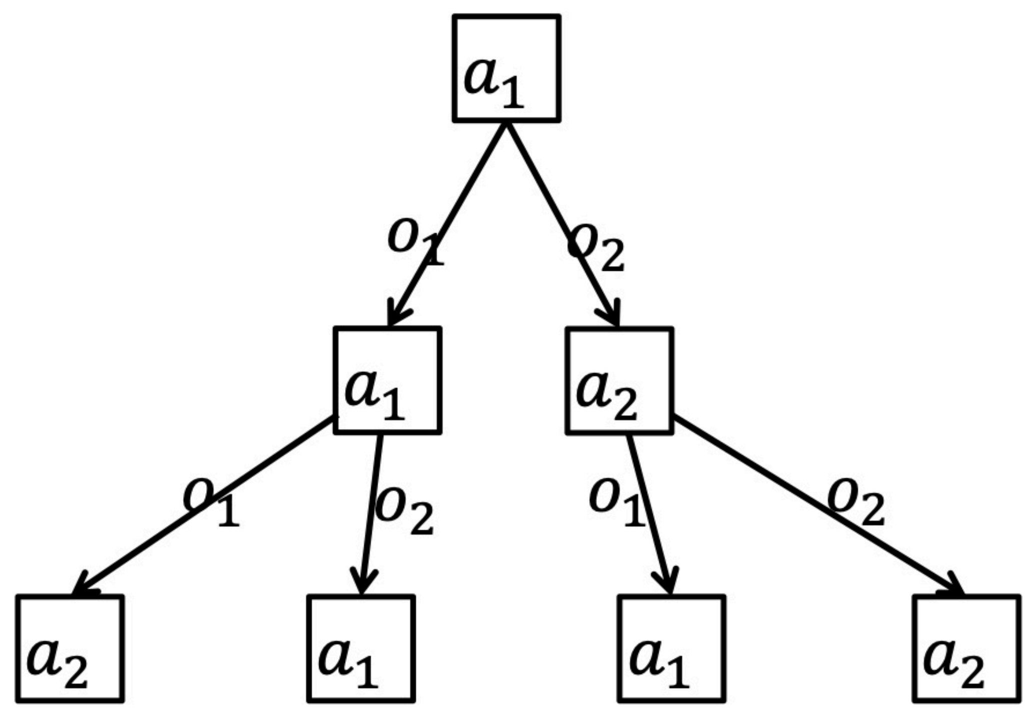
$$h_t \stackrel{\text{def}}{=} \langle a_0, o_1, a_1, o_2, \dots, a_{t-1}, o_t \rangle$$

  - Non-Markovian policy

Partially Observable Markov Decision Process (POMDP)

Policy Trees

- Policy $\pi: B \times H_t \rightarrow A_t$
- Consider a single initial belief b
- Then π can be represented by a tree **(composed of nodes)**



Belief updates

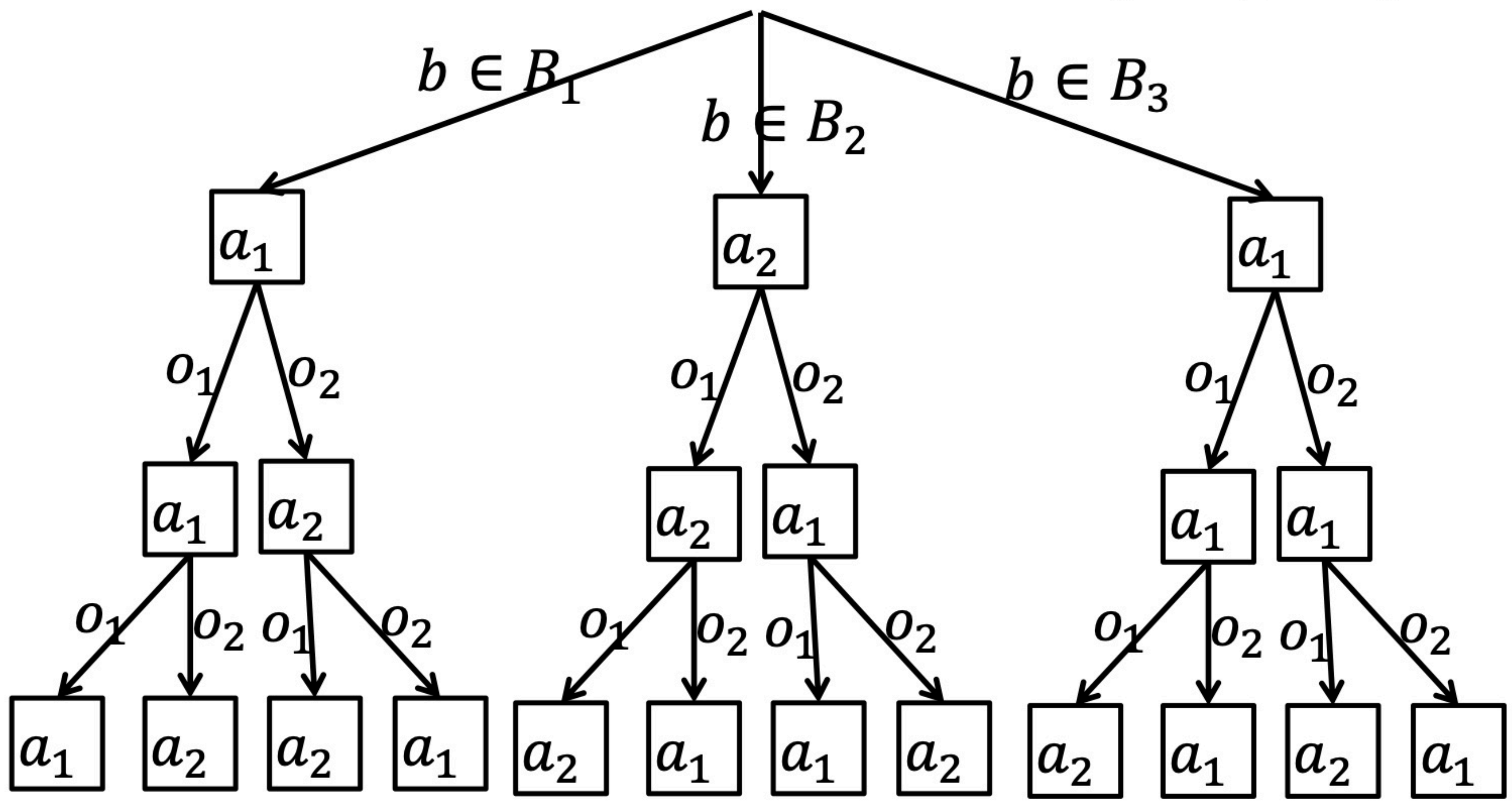
- **A policy tree is a decision-making approach representing the sequential decision-making process under uncertainty.**

Partially Observable Markov Decision Process (POMDP)

Policy Trees

- Policy $\pi: B \times H_t \rightarrow A_t$
  - Set of trees

Let $B = B_1 \cup B_2 \cup B_3$



Partially Observable Markov Decision Process (POMDP)

Beliefs

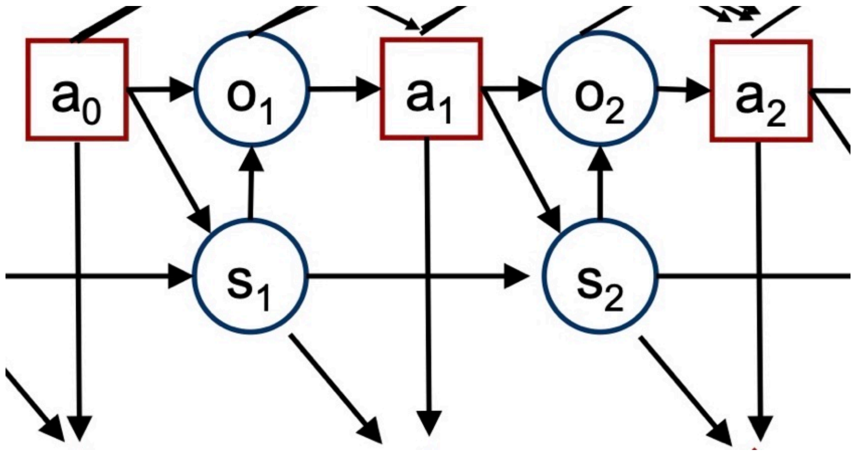
- Belief $b_t(s) = \Pr(s_t)$
  - Distribution over states at time t

- Belief about the underlying state based on history h_t **(POMDP)**

$$b_t(s) = \Pr(s_t | h_t, b_0)$$

Partially Observable Markov Decision Process (POMDP)

Belief Update



- Belief update: $b_t, a_t, o_{t+1} \rightarrow b_{t+1}$

$$b_{t+1}(s_{t+1}) = \Pr(s_{t+1} | h_{t+1}, b_0)$$

$$= \Pr(s_{t+1} | o_{t+1}, a_t, h_t, b_0) \qquad h_{t+1} \equiv o_{t+1}, a_t, h_t$$

$$= \Pr(s_{t+1} | o_{t+1}, a_t, b_t) \qquad b_t \equiv b_0, h_t$$

$$= \frac{\Pr(s_{t+1}, o_{t+1} | a_t, b_t)}{\Pr(o_{t+1} | a_t, b_t)} \qquad \text{Bayes' theorem}$$

$$= \frac{\Pr(o_{t+1} | s_{t+1}, a_t) \Pr(s_{t+1} | a_t, b_t)}{\Pr(o_{t+1} | a_t, b_t)} \qquad \text{chain rule}$$

$$= \frac{\Pr(o_{t+1} | s_{t+1}, a_t) \sum_{s_t} \Pr(s_{t+1} | s_t, a_t) b_t(s_t)}{\Pr(o_{t+1} | a_t, b_t)} \qquad \text{belief definition}$$

$$\propto \Pr(o_{t+1} | s_{t+1}, a_t) \sum_{s_t} \Pr(s_{t+1} | s_t, a_t)\, b_t(s_t)$$

Partially Observable Markov Decision Process (POMDP)

Markovian Policies

- Beliefs are sufficient statistics equivalent to histories (with the initial belief)

$$b_0, h_t \Leftrightarrow b_t$$

- Policies:
  - Based on histories: $\pi: B_0 \times H_t \rightarrow A_t$
    - Non-Markovian
  - Based on beliefs: $\pi: B \rightarrow A$
    - Markovian

Partially Observable Markov Decision Process (POMDP)

Belief State MDPs

- POMDPs can be viewed as belief state MDPs
  - States: B (beliefs)
  - Actions: A
  - Transitions:

$$\Pr(b_{t+1}|b_t, a_t) = \begin{cases} \Pr(o_{t+1}|b_t, a_t) & \text{if } b_t, a_t, o_{t+1} \rightarrow b_{t+1} \\ 0 & \text{otherwise} \end{cases}$$

  - Rewards: $R(b, a) = \sum_s b(s) R(s, a)$

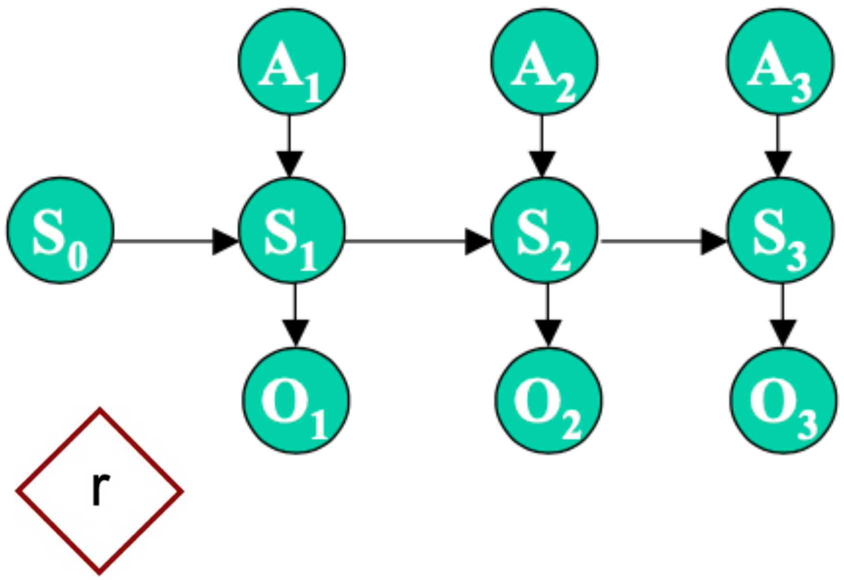
- Belief state MDPs
  - Fully observable
  - Continuous belief space

(belief state is a probability distribution over all possible states)

Partially Observable Markov Decision Process (POMDP)

POMDP ≡ Continuous-Space Belief MDP

- a POMDP can be seen as a continuous-space “belief MDP”, as the agent’s belief is encoded through a continuous “belief state”.



- We may solve this belief MDP like before using value iteration algorithm to find the optimal policy in a continuous space. However, some adaptations for this algorithm are needed.

Partially Observable Markov Decision Process (POMDP)

Policy Evaluation

- Value V^π of a POMDP policy π
  - Expected sum of rewards:

$$V^\pi(b) = E\left[\sum_t \gamma^t R(b_t, \pi(b_t))\right]$$

  - Policy evaluation: Bellman's equation

$$V^\pi(b) = R(b, \pi(b)) + \gamma \sum_{b'} \Pr(b'|b, \pi(b))\, V^\pi(b') \quad \forall b$$

  - Equivalent equation

$$V^\pi(b) = R\big(b, \pi(b)\big) + \gamma \sum_{o'} \Pr(o'|b, a)\, V^\pi(b^{a,o'}) \quad \forall b$$

Summary

- **Today we discussed**

  1- Introduction to POMDP

- **Next week**

  1- We will continue discussing the POMDP